

Problem

Consider the parallel RLC circuit in Fig. 8.60. What type of response will it produce?

- (a) overdamped
- (b) underdamped
- (c) critically damped
- (d) none of the above



Step-by-step solution

Step 1 of 5

The characteristic equation of a source free parallel RLC circuit is given as follows:

$$s^2 + \frac{1}{CR}s + \frac{1}{CL} = 0$$

The roots of the characteristic equation are:

$$s_{1,2} = -\frac{1}{2CR} \pm \sqrt{\left(\frac{1}{2CR}\right)^2 - \frac{1}{CL}}$$

The damping factor is,

$$\alpha = \frac{1}{2CR} \frac{\text{Np}}{\text{s}} \dots\dots (1)$$

The resonant frequency is,

$$\omega_0 = \frac{1}{\sqrt{CL}} \frac{\text{rad}}{\text{s}} \dots\dots (2)$$

Step 2 of 5

The source free parallel RLC circuit is,

Overdamped, if $\alpha > \omega_0$.

The source free parallel RLC circuit is,

Critically damped if $\alpha = \omega_0$.

The source free parallel RLC circuit is,

Under damped, if $\alpha < \omega_0$.

Step 3 of 5

Determine the damping factor of the circuit.

Substitute 1Ω for R , 1F for C and 1H for L in equation (1) .

$$\begin{aligned}\alpha &= \frac{1}{2 \cdot 1 \cdot 1} \\ &= \frac{1 \text{ Np}}{2 \text{ s}}\end{aligned}$$

Step 4 of 5

Determine the resonant frequency of the circuit.

Substitute 1Ω for R , 1F for C and 1H for L in equation (2) .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{1 \cdot 1}} \\ &= 1 \frac{\text{rad}}{\text{s}}\end{aligned}$$

Step 5 of 5

Determine the response of the circuit.

Compare both α and ω_0 .

$$\alpha = \frac{1 \text{ Np}}{2 \text{ s}} \text{ and } \omega_0 = 1 \frac{\text{rad}}{\text{s}} .$$

The system is not overdamped or critically damped because the damping factor is less than the resonant frequency.

Since $\alpha < \omega_0$, the response of the source free parallel RLC circuit is **underdamped** .